

Introduction to Machine Learning

Lab 1: Polynomial Regression

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1 Motivation

- Try to implement various training methods for polynomial regression.
- Implement commonly-used data normalization methods and check their influences on the mean-square-error (MSE) of testing data.

2 Tasks

1. Given the problem

$$\min_{\mathbf{w}} \|\mathbf{y} - \mathbf{X}\mathbf{w}\|_2^2, \quad (1)$$

where $\mathbf{X} = \{x_n^{d-1}\}_{n,d=1}^{N,D} \in \mathbb{R}^{N \times D}$ is the Vandermonde matrix, implement the function “training” to achieve its **closed-form** solution. **By setting different orders (i.e., D ’s), you can observe the change of the model.**

2. Given the problem in (1), implement the function “training_sgd” to learn the model via **stochastic gradient descent (SGD)**.
3. Given the above data matrix $\mathbf{X} = [\mathbf{x}_1, \dots, \mathbf{x}_D]$, implement different normalization strategies:

$$\tilde{\mathbf{X}} = \mathbf{X} \text{diag}(\mathbf{a}), \quad \mathbf{a} = \left[\frac{1}{\|\mathbf{x}_1\|_p}, \dots, \frac{1}{\|\mathbf{x}_D\|_p} \right]^T, \quad p = 1, 2, \infty, \quad (2)$$

where $\mathbf{a}$ is called scaling vector, and $\text{diag}(\cdot)$ means converting a vector to a diagonal matrix.

4. Achieve a SGD algorithm based on the normalized data and implement a testing method for the model trained on the normalized data.
5. Given the model parameters derived by training on the normalized data, derive the model parameters corresponding to the original unnormalized data directly. (Are they the same with those obtained in Task 1?)